

Arctic Consumer Behaviour Analysis

1. Project Overview: -

- Analyses customer shopping behaviour using transactional data from 3,900 purchases.
- Covers various product categories to guide strategic business decisions.
- Uncovers insights into spending patterns and customer segments.
- Examines product preferences and subscription behaviour.

2. Business Objectives: -

- A leading retail company wants to better understand its customers' shopping behaviour to improve **Sales, Customer Satisfaction, and Long-Term Loyalty**.
- The management team has noticed changes in purchasing patterns across **demographics, product categories, and sales channels (online vs. offline)**.
- They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.
- You are tasked with analysing the company's consumer behaviours dataset to answer the following overarching business question:
 - **“How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”**

3. Deliverables: -

- Data Preparation & Modelling (Python): Clean and transform the raw dataset for analysis.
- Data Analysis (SQL): Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
- Visualization & Insights (Tableau): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
- Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
- GitHub Repository: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

4. Dataset Summary: -

- Rows: 3,900
- Columns: 18
- Key Features: -
 - Customer demographics (Age, Gender, Subscription Status, etc.)
 - Purchase details (Item Purchased, Category, Purchase Amount, Size, etc.)
 - Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type.)
 - Missing Data: 37 values in Review Rating column.

5. Exploratory Data Analysis using Python: -

❖ We began with data preparation and cleaning in Python:

- Data Loading:

Imported the dataset using pandas.

- Initial Exploration:

Used df.info() to check structure and .describe() for summary statistics.

Summary statistics using .describe():-

```
df.describe(include = 'all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

Data Information:-

```
df.info()
✓ 0.0s Python

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Customer ID           3900 non-null   int64   
1   Age                   3900 non-null   int64   
2   Gender                3900 non-null   object  
3   Item Purchased        3900 non-null   object  
4   Category              3900 non-null   object  
5   Purchase Amount (USD) 3900 non-null   int64   
6   Location              3900 non-null   object  
7   Size                  3900 non-null   object  
8   Color                 3900 non-null   object  
9   Season                3900 non-null   object  
10  Review Rating          3863 non-null   float64  
11  Subscription Status    3900 non-null   object  
12  Shipping Type          3900 non-null   object  
13  Discount Applied       3900 non-null   object  
14  Promo Code Used        3900 non-null   object  
15  Previous Purchases     3900 non-null   int64   
16  Payment Method         3900 non-null   object  
17  Frequency of Purchases 3900 non-null   object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

- Missing Data Handling:

Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- Column Standardization:

Renamed columns to snake case for better readability and documentation.

- Feature Engineering:

- Created age_group column by binning customer ages.
- Created purchase_frequency_days column from purchase data.

- Data Consistency Check:

Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

6. Data Analysis using SQL (Business Transactions): -

We performed structured analysis in PostgreSQL to answer key business questions:

- Revenue By Gender:
Compared total revenue generated by **male vs. female customers**.

	gender	Revenue
▶	Male	157890
	Female	75191

- High-Spending Discount Users:
Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	61

➤ Top 5 Products by Rating:

Found products with the **highest average review ratings**.

	item_purchased	Average Product Rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

➤ Shipping Type Comparison:

Compared average purchase amounts between **Standard and Express shipping**.

	shipping_type	Average Purchase Amounts in %
▶	Express	60.48
	Standard	58.46

➤ Subscribers vs. Non-Subscribers:

Compared average spend and total revenue across subscription status.

	subscription_status	Total_Customers	Avg_Spend	Total_Revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

- Discount-Dependent Products:
Identified 5 products with the highest percentage of **discounted purchases**.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

- Customer Segmentation:
Classified customers into New, Returning, and Loyal segments based on **purchase history**.

	customer_segment	Number of Customers
▶	Loyal	3116
	Returning	701
	New	83

- Top 3 Products per Category:
Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

➤ Repeat Buyers & Subscriptions:

Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyers
▶	Yes	958
	No	2518

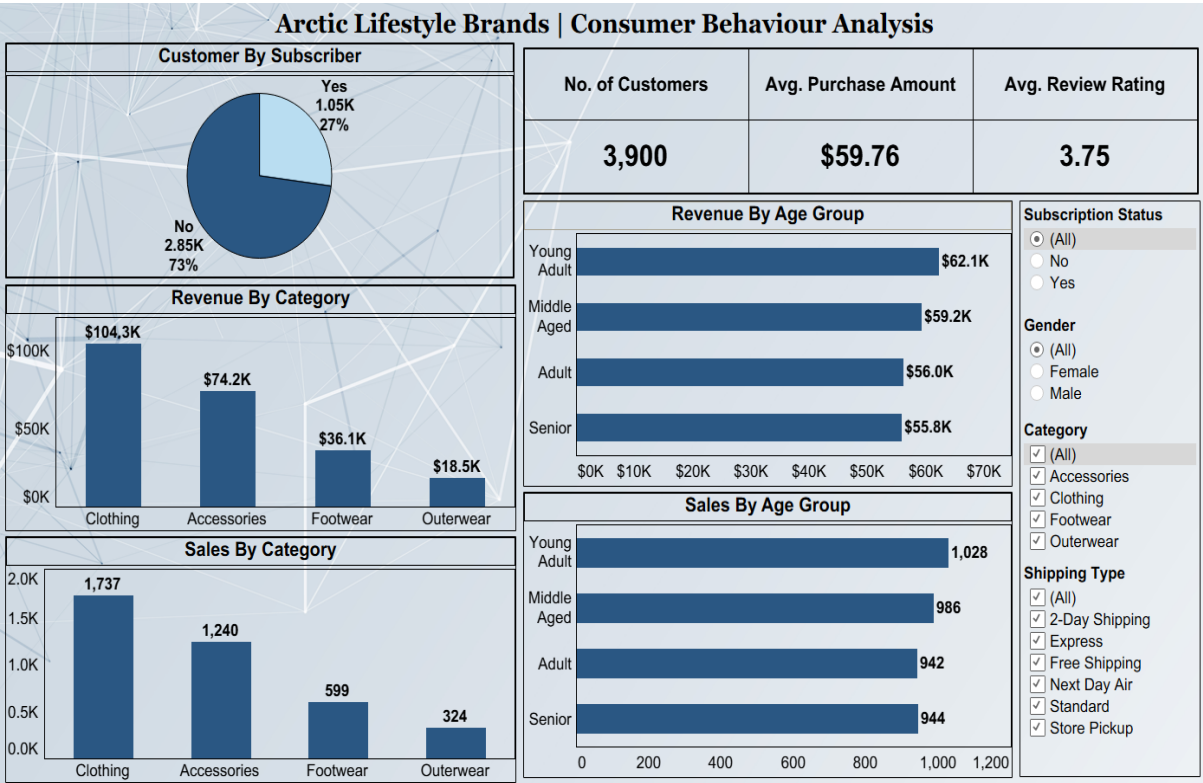
➤ Revenue by Age Group:

Calculated total revenue contribution of each **age group**.

	age_group	total_revenue
▶	Young Adult	62143
	Middle-Aged	59197
	Adult	55978
	Senior	55763

7. Dashboard in Tableau: -

Finally, we built an interactive dashboard in **Tableau** to present insights visually.



8. Business Recommendations:

- Enhance Subscription Value:
Lean into exclusive member benefits to drive **long-term recurring revenue**.
- Cultivate Brand Loyalty:
Implement rewards for frequent shoppers to transition them into **our high-value Loyal segment**.
- Optimize Discount Strategy:
Refine our promotional approach to ensure we are driving volume without compromising profit margins.
- Elevate Hero Products:
Feature our **highest-rated and top-selling** items more prominently in upcoming campaigns.
- Precision Marketing:
Direct our efforts toward our most profitable demographics and users who prioritize express shipping.